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Microstructure and deformation behaviour of austenitic low-density steels: The defining role of B2 intermetallic phase

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ABSTRACT

The feasibility of significant weight reduction in conjunction with the superior strength-ductility combination makes Fe-Mn-Al-C-based steels the candidate material for automotive and structural applications. The alloy chemistry and processing conditions influence the microstructure in low-density steels (LDS); however, their role in deformation behaviour is not fully understood. In the present study, three different austenitic LDS grades viz. Fe-28Mn-9Al-0.9C, Fe-28Mn-9Al-5Ni-0.9C and Fe-15Mn-9Al-5Ni-0.9C alloys in hot-rolled condition were used for evaluating the role of secondary phases, especially B2 ordered Fe(Ni)Al phase on deformation characteristics. Ni-free LDS microstructure consisted of the γ-matrix with fine κ-carbides, whereas Ni containing low Mn alloy possessed coarser κ-carbides along with B2 in both bamboo-like stringer and polygonal particle morphology in the γ-matrix. The B2 containing alloy exhibited higher strength with reduced tensile ductility than the B2-free alloy. Hardness of the γ-matrix and B2 phase were similar with B2 grains exhibiting distinct pop-in events in the nanoindentation curves indicating incipient plasticity. The γ-matrix and B2 co-deform near yield, and a favourable orientation relationship (OR) between the γ-matrix and B2 facilitated easy slip transfer, while the non-favourable OR controlled ductility by strain accumulation and B2-cracking. The differences in the strain hardening behaviour of B2-free and B2 containing alloys were elucidated based on the changes in dislocation substructure evolution examined by an automated crystal orientation mapping in electron microscopy.

1. Introduction

The demand for lightweight, high-strength structural materials has seen an exponential rise in recent times. The reduction in overall weight has become an essential criterion for several applications, especially in automotive and aerospace sectors, with a thrust to improve fuel efficiency and bring down emission levels. There have been significant developments in producing advanced high strength steels (AHSS) exhibiting a reasonable ductility necessary for forming various structural components. In this context, at present, there exists three generations of AHSS [1–3]. The 3rd generation AHSS are still under the nascent stage of research, whereas the 2nd generation AHSS have already gained importance to replace the 1st generation steels in specific structural components. Amongst the 2nd generation AHSS, Fe-Mn-Al-C based steel grades are popularly termed as either low-density steels (LDS) or lightweight steels with induced plasticity (L-IP). These alloys offer superior mechanical properties and extraordinary weight reductions because of alloy chemistry and complex strengthening mechanisms [4–6]. Based on the constitution of phases, LDS are classified as ferritic, austenitic, duplex, or triplex. Austenite (γ) based LDS alloys have four major microstructural features: (i) the γ-matrix, (ii) $L1_2$ ordered $(FeMn)_3AlC$ (κ-carbide) particles, (iii) B2 ordered Fe(Ni)Al type intermetallic compounds, and (iv) DO_3 ordered Fe_3Al precipitates [7–9]. The formation of these features depends entirely on alloy chemistry and thermomechanical processing conditions. Based on the size, shape, and site of occurrence, the morphology of B2 can be classified as (i) elongated cylindrical shapes called stringers which exist across several γ-grains, (ii) intergranular spherical (200 nm – 1 μm) at the recrystallised γ-grain boundaries, and (iii) intragranular precipitates (50 – 300 nm) formed within the shear bands in the deformed γ-grains [10]. It has also been reported that nickel addition would improve the strength of LDS alloys through the formation of Fe(Ni)Al type B2 phase [10].

Exhaustive literature is available on deformation behaviour and microstructural evolution in B2-free LDS [11–20]. However, limited studies exist which focus on understanding the influence of B2, with contrasting observations, on deformation behaviour of B2-containing LDS [21,22]. These studies involved the tensile deformation behaviour of cold-rolled plus annealed B2-containing LDS alloys. Yang et al.

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[21] have reported that B2 plays a significant role in deformation response. Strain incompatibility arises due to microstructural heterogeneity, leading to increased back stress in the γ-matrix. Both the back stress-induced hardening and forest dislocation hardening contribute to the high strain hardening ability of the alloy. On the contrary, Kim [22] showed experimentally and argued that B2 has a limited role in the deformation response of LDS alloys. It was reported that the γ-matrix composition plays a dominant role in the strain hardening. In addition, it was documented that the B2 phase in stringer morphology can affect forming by reducing ductility in transverse direction [23]. It is worth mentioning that the introduction of cold rolling followed by annealing heat treatment modifies the stringer orientation and its characteristics [10,24–26].

From the industrial processing point of view, the as-cast ingots/slabs undergo hot rolling practice before cold rolling (and annealing) to produce required sheets for forming any structural component. Thus, there is a necessity for understanding the effect of alloy chemistry on the microstructural evolution from casting to hot rolling and subsequently the role of the constituents on deformation behaviour of the hot-rolled LDS alloys before cold rolling. However, there are limited studies on deformation behaviour and the evolution of deformed microstructure in the hot-rolled LDS. This literature gap highlights the need for a systematic investigation to evaluate the alloy chemistry and the role of the resultant secondary phases such as κ-carbide, B2 stringers/particles on deformation behaviour of hot-rolled Fe-Mn-Al-C-(Ni) based LDS alloys.

In view of the aforementioned literature gaps, the present study was undertaken with specific objective of investigating the role of microstructural constituents, the B2 phase in particular, on the strength-ductility combination observed in the as-hot rolled plates of Ni-free and Ni containing LDS. Three low-density steel alloy compositions, which are known to produce B2-free and B2 containing γ-matrix, such as 28Mn0Ni, 28Mn5Ni, and 15Mn5Ni alloys with nearly constant Al and C contents, were melted, and the ingots were subsequently hot-rolled. The mechanical properties were evaluated through hardness (macro & nanoindentation) and tensile testing at room temperature on the hot-rolled alloys. The role of secondary phases on the deformation characteristics of LDS was deconvoluted by conducting detailed transmission electron microscopy (TEM) in conjunction with automated crystal orientation mapping (ACOM).

2. Materials and experimental procedure

2.1. Materials

For the present study, three different austenitic LDS grades were prepared by vacuum induction melting route. The defect-free portions of the 30 kg ingots were homogenised at 1423 K for two hours. The chemical composition, analysed by wet chemical method/standard emission spectroscopy, of the homogenised ingots are provided in Table 1. The three LDS grades: Fe-28Mn-9Al-0.9C, Fe-28Mn-9Al-5Ni-0.9C, and Fe-15Mn-9Al-5Ni-0.9C are hereafter referred to as 28Mn0Ni, 28Mn5Ni, and 15Mn5Ni, respectively. The concentration of Mn and Ni content were selected to obtain B2-free and B2-containing alloys based on stability of austenite. Two different contents of Mn and Ni each were selected based on the stability of austenite to produce B2-free and B2-containing microstructures. Hot rolling of the homogenised ingots was carried out at a temperature of 1323 K. Total thickness reduction of nearly 85% was achieved with ~15% reduction per pass.

Table 1
Chemical composition (in wt%) of LDS steels used in this study.

	Mn	Al	C	Ni	Fe
28Mn0Ni	28.3	9.0	0.9	–	Balance
28Mn5Ni	28.1	8.6	0.9	4.8	Balance
15Mn5Ni	14.7	9.8	0.9	4.1	Balance

2.2. Mechanical characterization

Macrohardness measurements were performed on the hot-rolled samples using Vickers indenter under a peak load of 300 N, as per ASTM E140–02 standards. The average of ten hardness readings taken at random locations on each sample was reported. Localised nanoindentation measurements were performed on the austenite matrix and B2 grains of the as-rolled samples at a peak load of 5 mN under constant loading/unloading rates of 500 μN s^{-1} and a dwell time of 5 s. To this end, EBSD is performed on a fiducially marked region followed by nanoindentation on select B2 and γ grains. The hardness and the reduced modulus of both phases were determined from load versus depth of penetration curves using the Oliver-Pharr model [27]. The average of measurements of nearly 50 grains of each phase was reported.

Cylindrical specimens with a gauge diameter of 6 mm and gauge length of 30 mm were prepared as per ASTM E8M standard for evaluating the tensile behaviour of the hot-rolled alloys. The tensile specimens were cut along the rolling direction (RD), with a few selected samples of 15Mn5Ni alloy cut along the transverse direction (TD) to investigate the in-plane tensile anisotropy. The tensile tests were carried out at room temperature and initial strain rate of 1×10^{-3} s^{-1} using LF200 Walter+Bai AG testing machine. The tensile elongation was measured using both punch marks and an extensometer on the gauge portion. Three samples were tested for each condition for reproducibility. Additionally, interrupted tensile tests were conducted on selected specimens to assess the microstructural evolution as a function of strain.

2.3. Microstructural characterization

Microstructural observations were carried out using an Olympus optical microscope. Electron channelling contrast imaging (ECCI) and orientation imaging microscopy (OIM) were carried out on an FEI scanning electron microscope (SEM) equipped with an EDAX electron backscattered diffraction (EBSD) detector. The samples were prepared by polishing on abrasive papers followed by final vibratory polishing with colloidal silica. Initially, large area EBSD scans of 500×500 μm^2 were obtained with a step size of 0.2 μm. Additional high-resolution scans were performed at an area of 100×100 μm^2 with a step size of 0.1 μm to resolve fine microstructural features. The OIM data was post-processed using EDAX TSL-OIM v8 software for evaluating grain orientations and phase fractions. The microstructures of the hot-rolled samples were examined along the RD-ND plane at the mid-section. The grain size was calculated using the linear intercept method with twin boundaries included during the analysis. Σ3 boundaries based on the coincidence site lattice (CSL) model were used to analyse twin boundary fractions in the microstructure. Twin boundaries were included in grain size calculations. The recrystallization fraction was estimated using the grain orientation spread (GOS) parameter. A critical value of 2° was chosen as the GOS cut-off for determining the recrystallised grain fraction. The composition of various phases was analysed by energy dispersive spectroscopy (EDS). Additionally, X-ray diffraction (XRD) was performed on the hot-rolled samples using cobalt K_α radiation at 135 mA current and 40 kV voltage in Rigaku smart lab X-ray diffractometer.

Transmission electron microscopy (TEM) studies were carried out on the hot-rolled and selected deformed tensile samples to examine the detailed microstructural features. Infrared thermography coupled with the tensile test was employed to identify the deformed region during tensile testing for TEM sample preparation. TEM sample preparation was initially done by mechanical polishing down to ~100 μm thickness. This was followed by twin jet electropolishing using an electrolyte consisting of nitric acid, hydrofluoric acid, sulfuric acid and lactic acid in methanol at 243 K at 18 V. Automated crystal orientation mapping (ACOM) was performed in TEM using Nanomegas ASTAR hardware and software

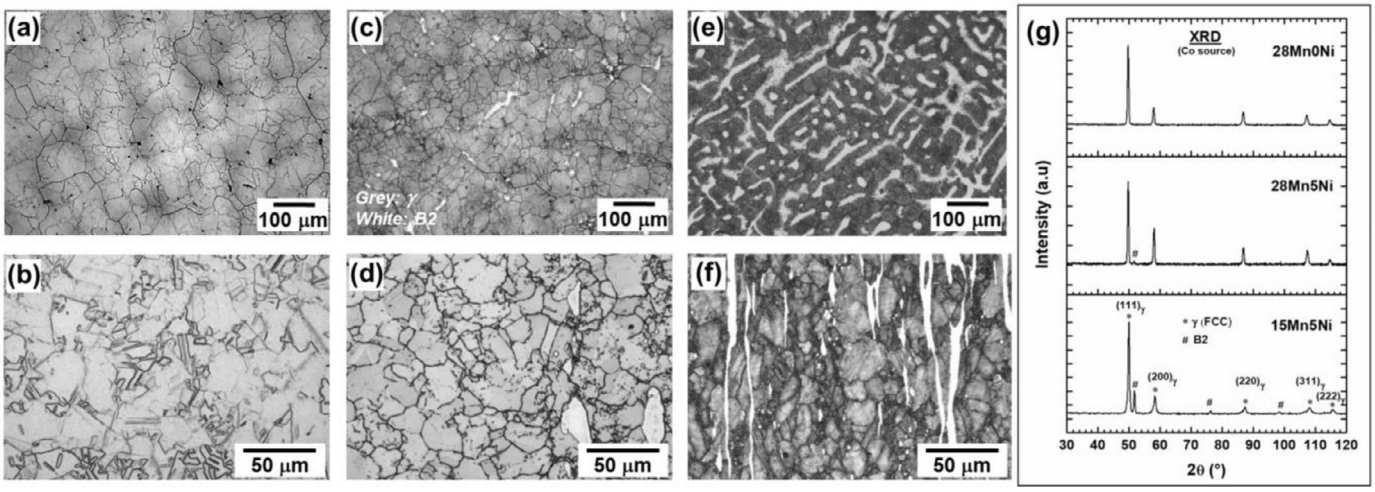


Fig. 1. Optical micrographs of as-cast plus homogenised and the hot-rolled (a, b) 28Mn0Ni, (c, d) 28Mn5Ni and (e, f) 15Mn5Ni alloys, respectively. (g) XRD patterns of the hot-rolled alloys.

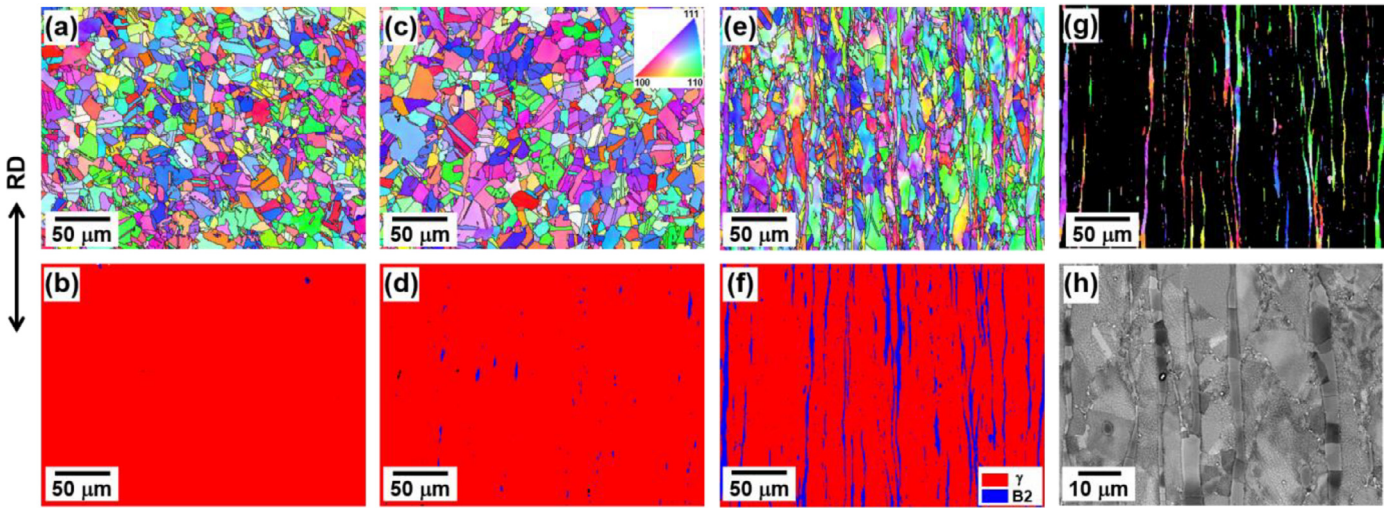


Fig. 2. Coloured OIM and the phase maps of the hot-rolled (a, b) 28Mn0Ni, (c, d) 28Mn5Ni and (e, f) 15Mn5Ni alloys, respectively. (g) IPF map of B2 stringers and (h) ECCI micrograph of the hot-rolled 15Mn5Ni sample at higher magnification.

packages. A precession angle of 0.5° was used for ACOM measurements, and the area of interest was scanned with a step size of 10 nm. The electron microscopy studies were performed on a Tecnai 20T G2 (FEI) TEM operated at 200 kV. In addition, fractography studies were carried out using a Zeiss EVO 18 SEM on the failed tensile specimens.

3. Results

3.1. As-homogenised cast and hot-rolled microstructures of LDS alloys

Fig. 1(a,b), 1(c,d), and 1(e,f) show the optical micrographs of 28Mn0Ni, 28Mn5Ni and 15Mn5Ni alloy samples, respectively in the as-cast plus homogenised and hot-rolled conditions. It is observed that the microstructure of the alloys containing Ni consists (28Mn5Ni and 15Mn5Ni) of two phases, whereas the 28Mn0Ni alloy is primarily made of a single phase. Fig. 1g presents the XRD patterns of the three alloys after hot rolling. Phase identification confirms the formation of B2 phase in γ-matrix in the Ni added alloys. In the hot-rolled alloys, the B2 phase exists as both polygonal particles and elongated stringers along the rolling direction. It is seen that the 15Mn5Ni alloy contains a relatively higher volume fraction of B2 phase than the 28Mn5Ni alloy. This suggests that the formation of the B2 phase with Ni addition is more favourable at lower Mn content. Further, it is observed that the B2 phase fraction slightly decreased upon hot rolling at 1323 K; for example, the 15Mn5Ni alloy showed a drop from ~15% in the as-cast condition to ~12% after hot-rolling. This decrease suggests a possible B2 dissolution under hot rolling conditions.

Fig. 2(a,b), 2(c,d), and 2(e,f) show the coloured OIM maps and the phase maps, respectively of the three alloys in hot-rolled condition. The grain sizes observed in the three alloys are found to be similar and are in the range of 5–7 μm after hot-rolling. Being medium SFE alloys [5,12], the hot-rolled microstructures are seen to contain a significant fraction of twin boundaries. The volume fraction of twins, V_f^T is found to be highest in the 28Mn5Ni (V_f^T ~40%) followed by 28Mn0Ni (V_f^T ~28%) and 15Mn5Ni (V_f^T ~7%). The OIM maps indicate that the crystallographic texture of the γ-matrix in the three alloys are similar and near-random. The volume fraction of B2 phase estimated from the phase maps is ~0.5%, 4% and 12%, respectively, in 28Mn0Ni, 28Mn5Ni, and 15Mn5Ni alloys. Fig. 2g illustrates the coloured OIM map for B2 stringers present in 15Mn5Ni alloy. The B2 stringers formed during hot-rolling are seen to be polycrystalline with a bamboo-like morphology. This observation is further corroborated with the ECCI micrograph shown in Fig. 2h of the hot-rolled 15Mn5Ni. The width of B2 stringers is seen to lie in the range of 1–5 μm. The elemental composition of B2 stringer and ma-

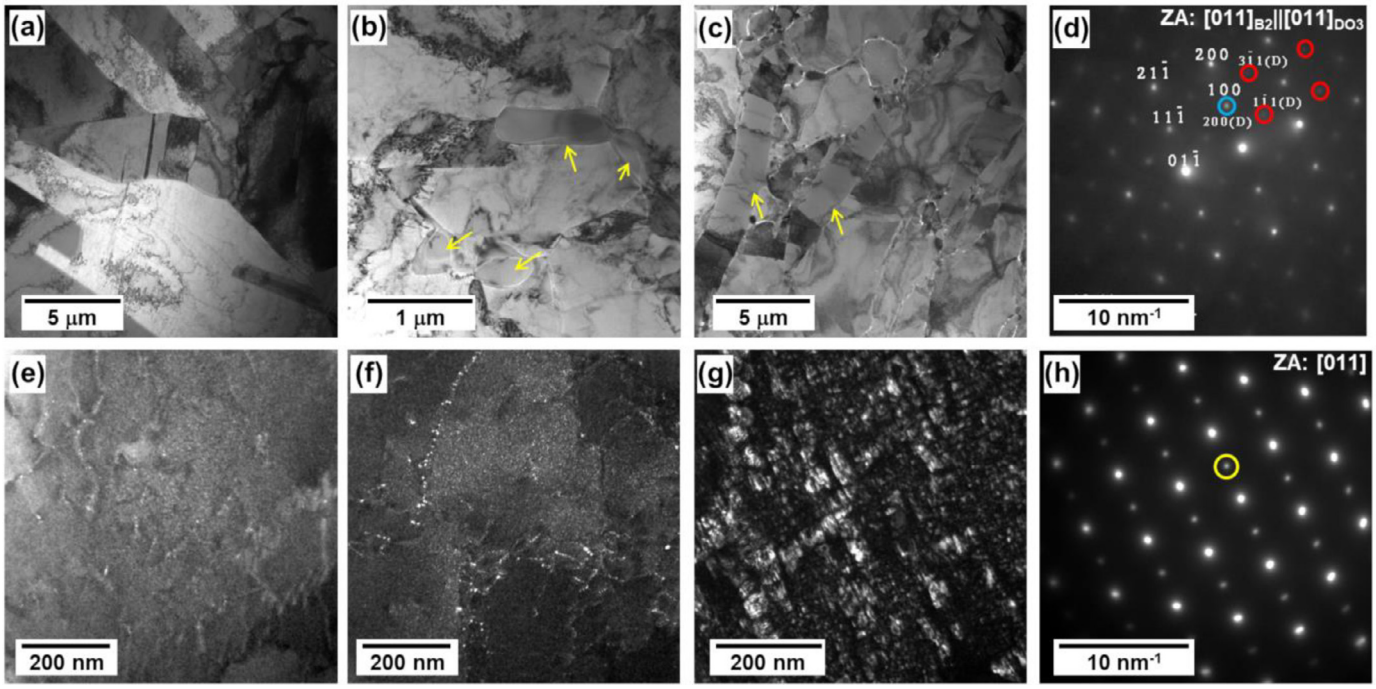


Fig. 3. Bright-field TEM images of the hot-rolled (a) 28Mn0Ni, (b) 28Mn5Ni and (c) 15Mn5Ni alloy samples. (d) SAED pattern from B2 grain indicating B2 and DO_3 reflections. Dark-field TEM images showing κ-carbide presence in (e) 28Mn0Ni, (f) 28Mn5Ni and (g) 15Mn5Ni, and (h) SAED pattern confirming κ-carbides presence.

trix was analysed using SEM-EDS and no compositional gradients were observed within B2 grains. However, it is noticed that the B2 stringers (Fe-11.3Mn-12.2Al-7.4Ni, in wt.%) are relatively richer in Al and Ni but deficient in Mn as compared to the γ-matrix (Fe-15.2Mn-9.1Al-4.7Ni, in wt.%) in the 15Mn5Ni alloy.

Fig. 3 shows the TEM micrographs of the hot-rolled 28Mn0Ni, 28Mn5Ni and 15Mn5Ni alloy samples. The bright-field images (Fig. 3a-c) confirm the microstructural features such as twins and B2 stringers in these alloys as noted earlier. The B2 particles having a size of ~300 nm are observed at γ-grain boundaries in 28Mn5Ni alloy (as marked by yellow arrows in Fig. 3b). Fig. 3c shows large B2 stringers connected across multiple γ-grains along with finer B2 at γ grain boundaries in the 15Mn5Ni alloy (marked with yellow arrows). Fig. 3d shows the selected area electron diffraction (SAED) pattern taken from one of the stringers along [011] zone axis, which confirms it to be an ordered B2 phase. It is also observed that there were secondary spots visible in the SAED pattern apart from the B2 spots. This suggests an additional phase present in B2 and could be related to the DO_3 phase, as reported earlier [28]. An orientation relationship of $[011]_{B2}||[011]_{DO3}$ is found to exist between the two phases. Differences in composition (as analysed using TEM-EDS) are observed between the B2 phase present in 28Mn5Ni and 15Mn5Ni alloys. The composition of B2 in 15Mn5Ni alloy is Fe-10.1Mn-9.7Al-8.3Ni, whereas in the 28Mn5Ni alloy, it is Fe-16.3Mn-13.6Al-21.5Ni (wt.%). Thus, the B2 forming alloys showed a general trend of showing higher Ni, Al and Mn concentrations in the B2 phase with increasing Mn content of the alloy. Figs. 3e-g show the dark-field TEM images, highlighting κ-carbide particles in the three alloys. The images were obtained from the weak diffraction spot encircled in the SAED pattern taken along the $[011]_{\gamma}$ zone axis, as shown in Fig. 3h, for selected austenite grains of the three alloys. The presence of fine κ-carbide particles in all three alloys is consistent with earlier reports [29,30]. The dark-field image of 15Mn5Ni alloy (Fig. 3g) shows the formation of cuboidal-shaped κ-carbides, having a size in the range of 10 to 40 nm, consistent with the earlier report by Hwang et al. [28]. Furthermore, the dark-field images (Fig. 3e-g) reveal that the κ-carbides occur within the matrix grain and along grain boundaries. The volume fraction of κ-carbides is highest in 15Mn5Ni alloy followed by 28Mn5Ni and 28Mn0Ni alloys. This indicates that Ni addition favours the formation of κ-carbides at constant Mn content. Also, it can be noted that increasing the concentration of Mn results in suppressing the formation of κ-carbides. This may result from modification of the phase diagram or transformation curve so that more κ-carbides are formed in 15Mn5Ni alloy during cooling after hot rolling. The aforementioned microstructural features (grain size, twin boundary and B2 volume fraction) and the elemental analysis using TEM-EDS of B2 particles/stringers and the adjacent γ-matrix grains of all the three alloys are summarised in Table 2. EDS elemental analysis in both SEM and TEM suggests a clear solute partitioning existing between γ and B2 phases in the LDS alloy.

3.2. Mechanical behaviour of hot-rolled LDS alloys

Macrohardness measurements, summarised in Table 3, indicated that 15Mn5Ni alloy exhibits higher average hardness (HV375) than the remaining two LDS alloys. Nickel addition has a marginal effect on hardness as both 28Mn0Ni and 28Mn5Ni alloys display hardness of HV256 and HV268, respectively.

Fig. 4a,b show the load-depth of penetration curves (10 for each phase) obtained from localised nanoindentation of the γ-matrix and polycrystalline B2 stringers, respectively in 15Mn5Ni alloy. The selected regions of interest for nanoindentation presented as coloured OIM maps are shown in Fig. 4c-e and as phase+IQ maps in Fig. 4f-h. The ECCI micrographs (Fig. 4i-k) reveal the indents on the individual γ and B2 stringer grains. Fig. 4c,f,i are of the γ-matrix grains, while the rest highlight B2 stringer grains. The average hardness values of the γ-matrix and B2 phase are listed in Table 3. Interestingly, both the γ-matrix and B2 phase exhibit nearly similar hardness values of 6.7–6.8 GPa and the reduced elastic moduli of 169–172 GPa. These observations are in close agreement with the recent results reported by Rahnama et al. [25,26] in steels of similar composition. Furthermore, the nanoindentation curves of B2 grains showed the occurrence of pop-in(s) events at low loads during loading. The enlarged view of the pop-in(s) is shown as an inset in Fig. 4b.

Table 2
Microstructural features observed in the three alloys under hot-rolled conditions.

Alloy	Phases present	Phase fraction of B2 (%)	Average grain size (μm)	Twin fraction (%)	Composition of phases (wt%)
28Mn0Ni	$\gamma + \kappa$	~0.5	5.6 ± 2.3	28	γ: Fe-27.5Mn-7.9Al
28Mn5Ni	$\gamma + \kappa$ + intergranular B2 particles +DO_3	~4	6.9 ± 1.7	40	γ: Fe-27.8Mn-7.5Al-5.7Ni B2: Fe-16.3Mn-13.6 Al-21.5Ni
15Mn5Ni	$\gamma + \kappa$ + intergranular B2 particles+ B2 stringers + DO_3	~12	5.7 ± 1.3	07	γ: Fe-14Mn-7.6Al-4.9Ni B2: Fe-10.1Mn-9.7 Al-8.3Ni

Table 3
Hardness and tensile properties of the three alloys in hot-rolled condition.

Alloy	Tensile Direction	Macro-hardness (VHN)	Hardness (Nanoindentation) (GPa)	YS (MPa)	UTS (MPa)	Elongation (%)
28Mn0Ni	RD	256 ± 5	–	670 ± 12	980 ± 10	68 ± 3
28Mn5Ni	RD	268 ± 5	–	730 ± 20	1028 ± 20	55 ± 3
15Mn5Ni	RD		γ: 6.7 ± 0.5	1110 ± 8	1260 ± 5	20 ± 2
	TD	375 ± 6	B2: 6.8 ± 0.3	1150 ± 5	1261 ± 10	16 ± 2

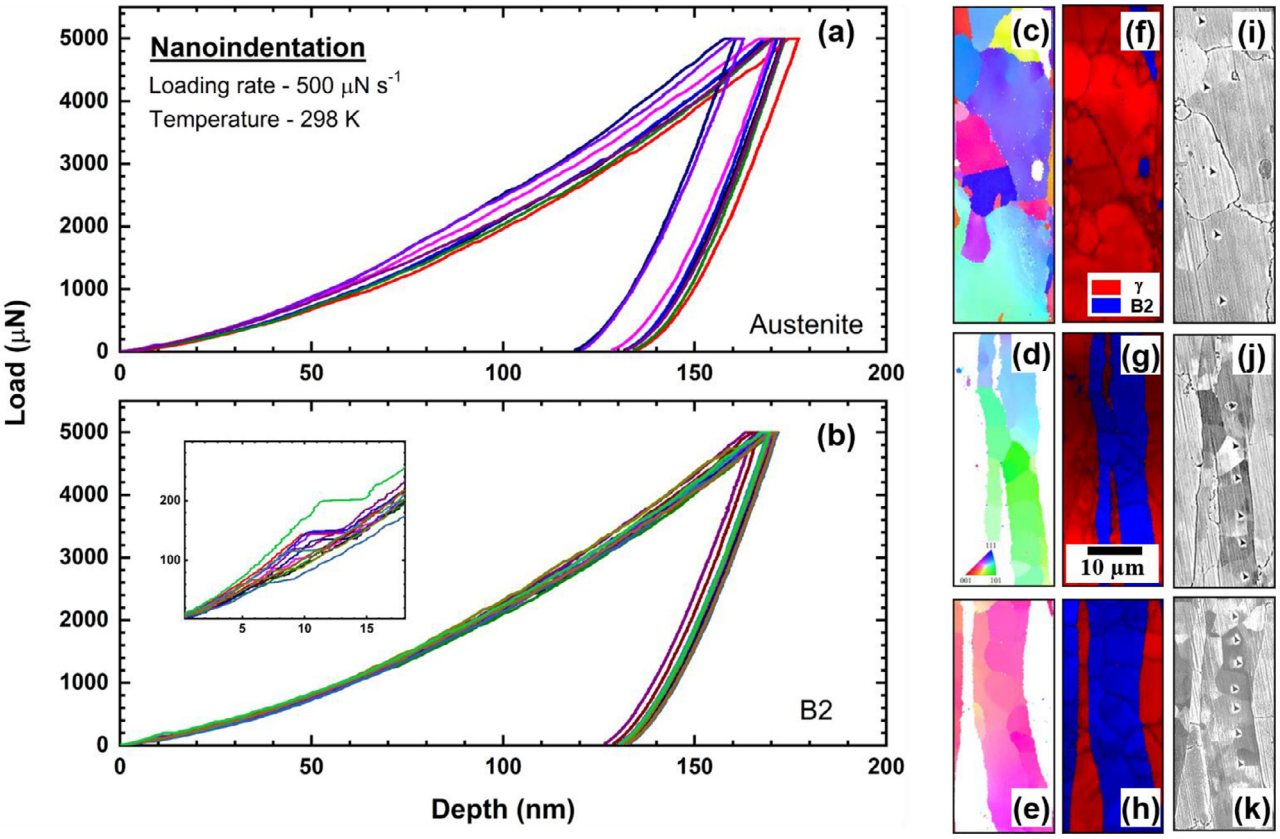


Fig. 4. (a,b) The load-depth curves of γ and B2 phase, (c,f,i) OIM, phase and ECCI maps of the indented γ-matrix and (d,g,i) and (e,h,k) show indented regions of two B2 stringers in 15Mn5Ni alloy respectively.

Fig. 5 illustrates the tensile response of the three alloys in the hot-rolled condition. The flow curves of the three different alloy specimens along the rolling direction are shown in Fig. 5a. The 15Mn5Ni alloy exhibited significantly higher yield and tensile strengths and reduced ductility than the other two LDS alloys. The differences observed in the tensile properties of 28Mn0Ni and 28Mn5Ni alloys were minor. The Ni containing alloy exhibited slightly higher strength and lesser ductility. As the hot-rolled 15Mn5Ni alloys' microstructure showed coarse B2 stringers aligned along RD, tensile tests were carried out in the transverse direction (TD) as well to ascertain their role on in-plane tensile anisotropy. Table 3 summarises the tensile properties such as the yield strength (YS), ultimate tensile strength (UTS) and percentage of elongation to failure of LDS. The flow curves along RD and TD direction (Fig. 5a) were almost similar, indicating minimal tensile anisotropy after hot-rolling. The magnified portion near yield in the tensile curves of 15Mn5Ni alloys in the RD and TD directions is shown as an inset in Fig. 5a. There were no noticeable plateaus. Park et al. [23] reported similar observations, where LDS alloys containing coarse B2 stringer bands did not show change in strength values along RD and TD. The observed reduction in ductility in the rolling and transverse directions can be attributed to the lack of favourable grain orientation on slip transfer at γ/B2 interface. The role of the B2 phase, κ-carbides, and the dislocation

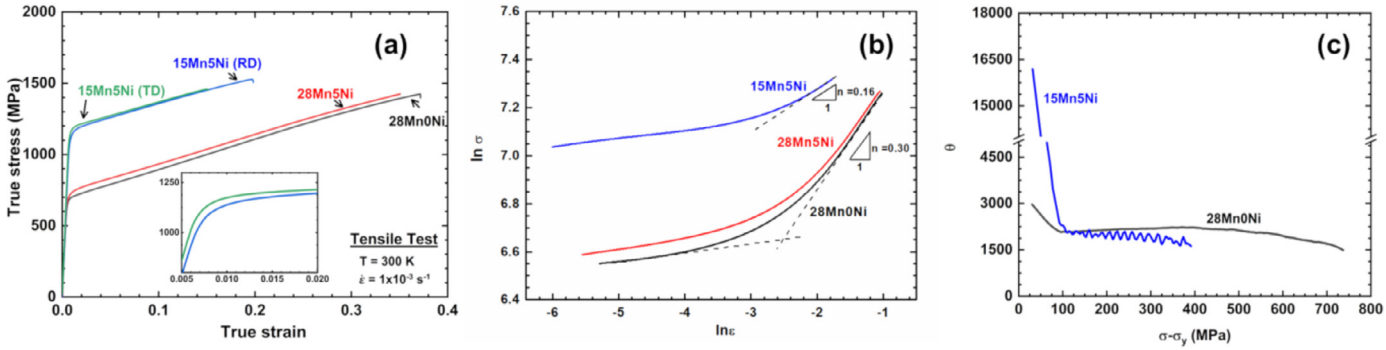


Fig. 5. (a) True stress-true strain curves and (b) logarithmic plot of true stress - true strain curves of 28Mn0Ni, 28Mn5Ni and 15Mn5Ni alloys, and (c) strain hardening rate-stress curves of 28Mn0Ni and 15Mn5Ni alloys.

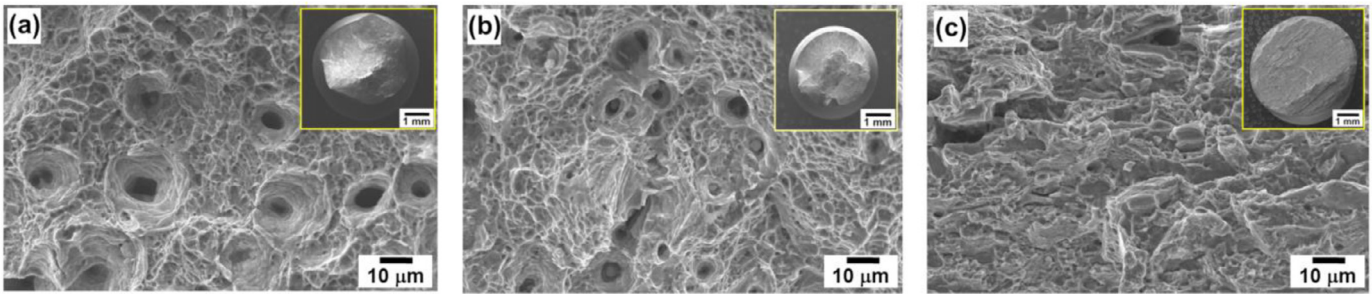


Fig. 6. Fractographs of (a) 28Mn0Ni, (b) 28Mn5Ni and (c) 15Mn5Ni alloys specimens failed in tension. Insets are low magnification SEM images of one half of the tensile specimens after failure.

substructure evolution on the mechanical response of these LDS alloys are discussed later.

The strain hardening response was discerned by analysing the stress-strain data to determine the strain hardening exponent (n) and the strain hardening rate (θ). Fig. 5b presents the plot of $\ln(\sigma)$ versus $\ln(\varepsilon)$ for the three flow curves (along RD) as shown in Fig. 5a. It is noted that there are two slopes in the linear curves corresponding to two different strain hardening exponents (n) with strain. At low strains, the n value is relatively small and is independent of the alloy composition. However, the n value at higher tensile strains is significantly higher for both 28Mn0Ni and 28Mn5Ni alloys (n~0.30) than that of 15Mn5Ni alloy (n~0.16) and is consistent with an earlier report [22]. The differences in strain hardening are related to the presence of κ-carbide and B2 phases in microstructure, which will be discussed later. The variation in strain hardening rate with stress increment after yield ($\sigma - \sigma_y$) for two alloys (28Mn0Ni and 15Mn5Ni) is shown in Fig. 5c. The 15Mn5Ni alloy initially exhibited a sharp fall in stage III strain hardening followed by a constant strain hardening rate. The 28Mn0Ni alloy also showed a similar trend, although there was a slight increasing strain hardening rate over a significant range of stress (strain) after stage III hardening rate, and subsequently the strain hardening rate decreased. The strain hardening behaviour of 28Mn5Ni alloy, was found to be similar to that of 28Mn0Ni alloy and not shown here for the sake of brevity.

3.3. Post-tensile deformed microstructures

Fig. 6 shows the fracture surfaces of the specimens of the three alloys after tensile testing. The 15Mn5Ni alloy showed a distinctive fracture surface topography from that of the other two alloys. The insets in each image show one-half of the failed tensile specimens. Both 28Mn0Ni and 28Mn5Ni alloys exhibit clear cup and cone type failure (insets), as evident from the dimples. This suggests ductile failure involving microvoid formation, growth and coalescence. The 15Mn5Ni alloy specimen displayed numerous micro-cleavage facets connected by dimples on a nearly flat surface at the macroscale (Fig. 6c). This suggests that the fracture in 15Mn5Ni alloy is crack initiation-controlled, triggered by the B2 stringers. Further microstructural analysis was done to ascertain the role of B2 on the strength of the alloy and changes in the fracture behaviour.

Transmission electron microscopy was performed on 28Mn0Ni and 15Mn5Ni alloy samples to deconvolute the effect of microstructural constituents on yield strength, strain hardening and ductility. Specimens were deformed to true strains of ~0.015 and ~0.15 in tension, representing the two different strain hardening regimes observed in the alloys (Fig. 5b). Fig. 7 shows TEM micrographs of 28Mn0Ni and 15Mn5Ni alloys deformed to ~0.015 strain. Fig. 7a shows dislocation pinning in the deformed 28Mn0Ni alloy. For the same applied strain, the 15Mn5Ni alloy showed significant dislocation accumulation in the austenite channels in between the κ-carbide particles, as seen in Fig. 7b. Both the γ-matrix and B2 stringers underwent significant deformation as evident from the dislocation activity. An important observation is the slip transfer that occurs at the interface of austenite and B2 stringer marked with arrows in Fig. 7c. Slip transfer would suggest that there exists substantial load sharing between the phases. The SAED pattern (inset of Fig. 7c) shows a specific orientation relationship (OR) existing between the austenite and B2 stringer grains. However, slip transfer was not observed across all γ/B2 interfaces with respect to a single B2 stringer and surrounding γ-matrix, as seen in Fig. 7d. This would imply that the γ-matrix and the grains within B2 stringers can deform simultaneously near yield. This corroborates the localised nanoindentation measurements (Fig. 4, Table 3) performed on B2 grains and γ-grains.

To further evaluate the observed slip transfer activity, the 15Mn5Ni alloy specimen deformed in tension (ε~0.015) was examined using automated crystal orientation mapping (ACOM) in TEM. About ten such regions were examined. Fig. 7e-h presents the ACOM results from a deformed region containing both γ-grains and B2 particle/stringer in the tensile specimen. The virtual bright-field image, OIM micrograph, the phase map, and kernel average misorientation (KAM) maps of the

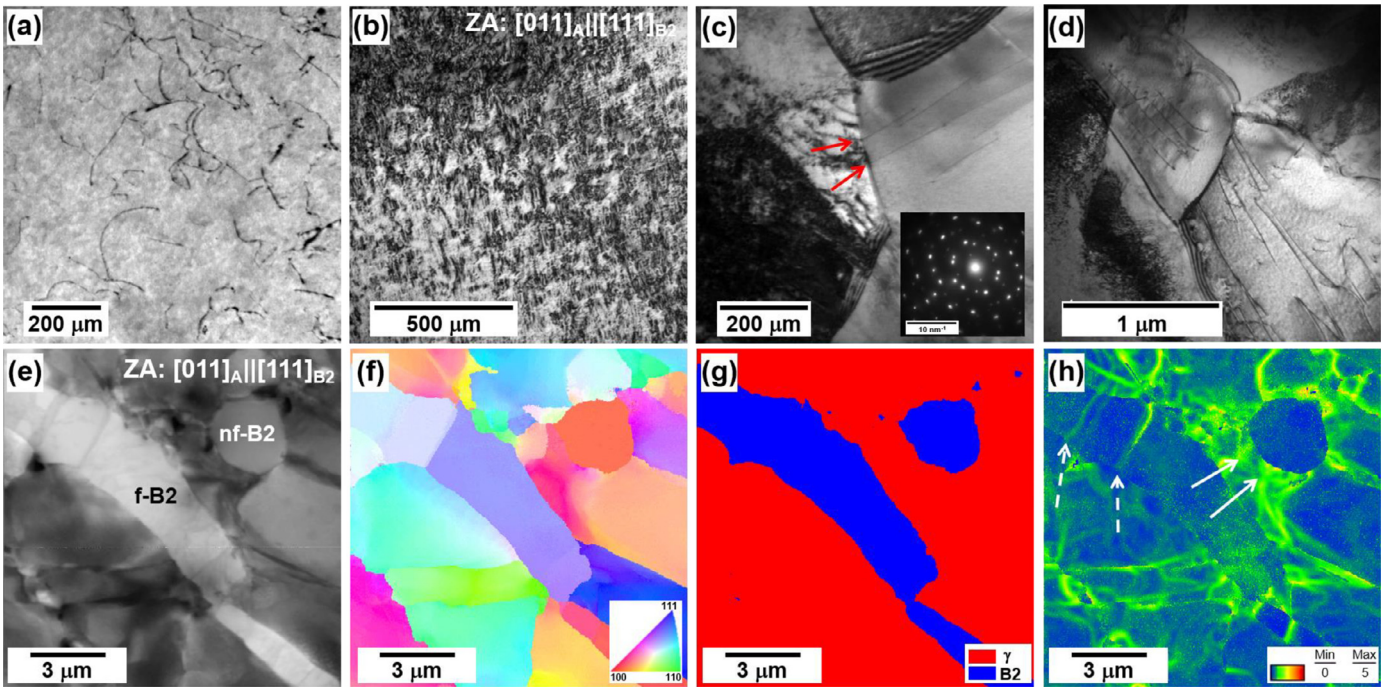


Fig. 7. Bright-field TEM micrographs showing dislocation interaction with κ-carbides in samples deformed near yield point of (a) 28Mn0Ni and (b) 15Mn5Ni alloys, (c) slip transfer at γ/B2 interface having favourable OR between γ and B2, (d) slip restriction at B2/B2 interface. (e) ACOM virtual bright-field, (f) coloured orientation map, (g) phase map, and (h) KAM map showing strain distribution inside the phases and at interfaces.

selected region are shown in Fig. 7e-h, respectively. The KAM map (Fig. 7h) highlights traces of slip transfer across interfaces of the B2 stringer (named 'f-B2' in Fig. 7e) and matrix grains where the OR exists. However, substantial strain accumulation was observed at the interfaces where OR is not present (named 'nf-B2' in Fig. 7e) i.e. having hard orientations. The KAM maps also corroborate the TEM observations of strain accumulation around B2 grains but no slip transfer between them. The slip transfer across f-B2 is marked with dashed arrows, whereas the strain accumulation around nf-B2 is highlighted with solid arrows in the KAM map shown in Fig 7h.

Fig. 8 shows TEM micrographs of 28Mn0Ni and 15Mn5Ni alloys deformed to a true strain of ~0.15. There was significant slip band formation as shown in Fig. 8a in the γ-grains of 28Mn0Ni alloy. The interaction of these slip bands with annealing twins was also observed, as shown in Fig. 8b. In the case of the 15Mn5Ni alloy, after a substantial accumulation of dislocations within the γ-grains at low strain (ε~0.015), the κ-carbide has undergone shearing by slip activity upon further straining (Fig. 8c). The presence of these particles can also affect subsequent dislocation substructure evolution in γ-matrix. The deformed microstructure in these alloys was characterised by pronounced planar dislocation glide (Fig. 8a and d) where slip bands are primarily crystallographic glide planes having high dislocation density. This observation is seen to be consistent with previous reports [17,18,20,31]. Although both 28Mn0Ni and 15Mn5Ni alloys show the formation of slip bands, the average spacing between the bands was observed to be much finer in 28Mn0Ni (~100–200 nm) as compared to in 15Mn5Ni alloy (~400–600 nm). Moreover, the slip band spacing was found to be relatively inhomogeneous in 15Mn5Ni alloy. Also, significant dislocations pile-up was noticed around B2 particles (Fig. 8e), which was observed earlier at a lower tensile strain. The 15Mn5Ni alloy deformed to ~0.15 strain exhibited sparse deformation twins, highlighted by arrows in Fig. 8f, which were absent in the 28Mn0Ni alloy.

The slip transfer activity occurring at γ/B2 interface in the deformed LDS specimens (ε~0.15) was further analysed using ACOM with multiple grains analysed for statistical relevance. The ACOM image quality (IQ) and phase maps of selected regions in 28Mn0Ni and 15Mn5Ni alloys are shown in Fig. 9a,b and c,d respectively. Broken arrows in Fig. 9a and c mark prominent slip band traces present in the γ-grains of 28Mn0Ni and 15Mn5Ni alloys. Point-to-point misorientation profiles in deformed γ-grains of 28Mn0Ni (along A-A' line) and 15Mn5Ni alloys (along B-B' line) are presented in Fig. 9e. This observation confirms fine deformation twin formation with misorientation angles of ~55–60° in 15Mn5Ni alloy. The KAM map also reveals increased strain localization around fine B2 particles and stringers in γ-matrix grains of 15Mn5Ni alloy (Fig. 9f). During deformation, fine B2 at the γ-grain boundaries can restrict slip leading to strain localization and thereby enable sufficient back stress, which initiate deformation twinning in selected austenite grains.

To understand the role of B2 stringer on ductility and fracture behaviour of 15Mn5Ni alloy, the longitudinal section of the failed tensile specimens was examined using SEM-EBSD. The IQ map of the region is shown in Fig. 10a. It is observed that some B2 stringers have cracked in a direction perpendicular to the tensile axis at specific locations (identified by arrows in Fig. 10a). The OIM results (IPF, phase and KAM maps) obtained from a selected region, marked by a rectangular box in Fig. 10a, are presented in Fig. 10b-d. It is noted that there is substantial strain accumulation and cracking of some deformed B2 grains of specific orientations. Fig. 10e illustrates the pole figures of B2 stringer and adjacent γ-grain for two selected regions shown in Fig. 10b. Region-I represents a case where the B2 stringer cracked during tensile testing and region-II shows a non-cracked B2 stringer. It is noteworthy that cracks formation was observed in the B2 stringers which exhibit no OR with austenite grain. This observation suggests that the fracture of B2 stringers due to restricted slip transfer at the γ/B2 interface can adversely affect the overall ductility of LDS alloys.

4. Discussion

4.1. Effect of Mn and Ni content on evolution of secondary phases in LDS alloys

The physical metallurgy aspects of LDS are quite complex, as each elemental alloying addition has a specific role and exhibits a synergistic effect on evolution of microstructure and properties. The addition

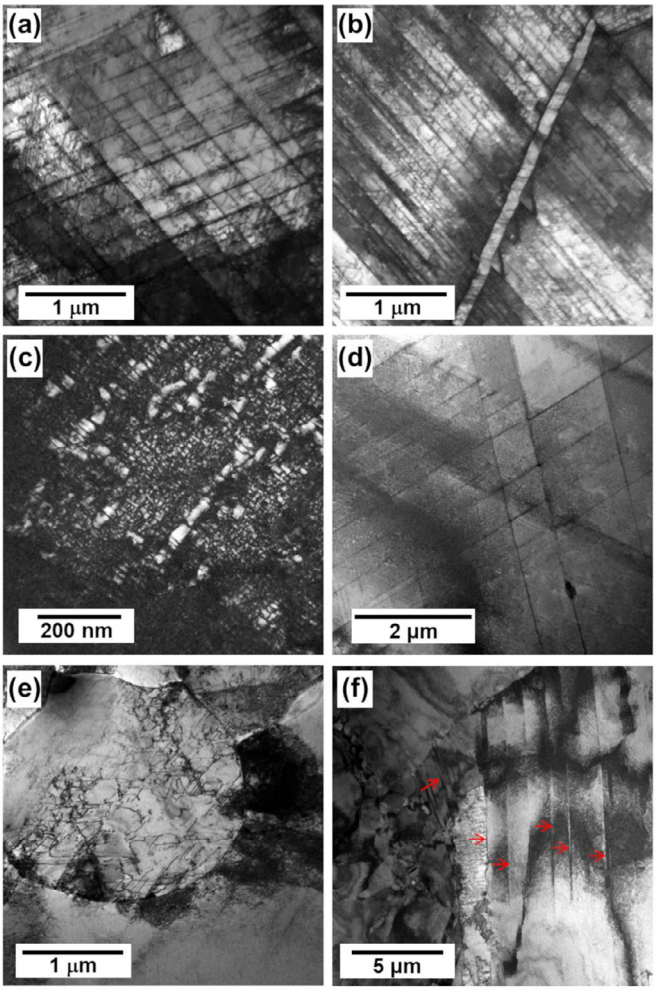


Fig. 8. TEM micrographs showing (a) slip bands and (b) slip band-annealing twin interaction in 28Mn0Ni alloy, whereas (c) shearing of κ-carbides, (d) slip band, (e) dislocation pile up around B2, and (f) deformation twins in few γ-grains in 15Mn5Ni alloy. The samples were deformed to a tensile strain of 0.15.

of high amounts of strong austenitic stabilisers such as C, Mn and Ni can result in a fully austenitic matrix at room temperature. Chang et al. [32] showed that with increasing the carbon content from 0.26 to 1.06 wt%, there would be a reduction in the volume fraction of ferrite from ~55 to 0% i.e. changing from duplex to γ-matrix LDS. The hot-rolled microstructure of Fe-28Mn-9Al-0.9C alloy in the present study was comparable to the one reported by Etienne et al. [33] for a similar LDS composition. In the current study, a comparison between the microstructures of the hot-rolled 28Mn5Ni and 15Mn5Ni alloys shows a significant reduction in the volume fraction of B2 stringers/particles with an increase in Mn levels for the same Ni content. This observation was consistent with those reported by Hwang et al. [28] where a B2 stringer-free microstructure was obtained in Fe-21Mn-10Al-1C-5Ni alloy due to the high Mn and C contents. As anticipated, the presence of higher Mn and C contents stabilised the austenite matrix in the alloys. However, it seems Al played increased stability of the ferrite phase and also formation of an ordered B2 phase under the current processing conditions. Moreover, Ni addition is known to lower the enthalpy of ordered phase and thereby, enhances the stability domain of ordered B2 phase [10]. Higher volume fractions of secondary phases such as κ-carbides, B2 and DO_3 ordered intermetallic compounds can be expected in 15Mn5Ni alloy because of the lower stability of austenite in the low Mn LDS grades. The B2 phase, which exhibits similar hardness levels as that of the γ-matrix, is expected to deform and elongate along rolling direction resulting in the formation of B2 stringers. The thermo-mechanical cycle can significantly affect the morphology of the B2 phase, which has scope for a separate study. It is worth mentioning that κ-carbide precipitation in Fe-Mn-Al-C based alloy systems is found to be influenced by chemical composition, processing conditions, cooling rate, etc., [5, 12-13, 15, 18, 20-21]. Frommeyer and Brüx [5] reported a triplex structure in Fe-26Mn-11Al-1.1C consisting of $\gamma + \alpha$ (5%) + κ-carbides. Their study involved on the alloy subjected to cold rolling followed by recrystallization at 1323 K. The presence of κ-carbides was observed in the samples processed well above the typical ageing temperature (773–923 K). A separate study by Sutou et al. [13] reported the formation of fine κ-carbides in Fe-20Mn-Al-C-Cr alloys heat treated at 1373 K. In their study, very fine precipitates were observed in the water-quenched samples, while the air-cooled samples contained coarser cuboidal κ-carbide precipitates. This suggests that mode of cooling and processing condition can affect the elemental partitioning which may influence the nature of κ-carbide precipitates. In the present study involving similar processing conditions, finer κ-carbides are seen in both 28Mn0Ni and 28Mn5Ni alloys, whereas the 15Mn5Ni alloy exhibited relatively coarser κ-carbides in cuboidal morphology. This could primarily be due to the differences in elemental partitioning because of the changes in alloy composition dependent stability of austenite phase. Nevertheless, this demands a detailed study on evolution of various phases and their morphology as a function of alloying content and processing conditions using thermodynamic and kinetics modelling and it is beyond the scope of the present study.

4.2. Role of secondary phase on deformation behaviour of LDS alloys

The mechanical (indentation and tensile) test results revealed that the hardness, strength, and ductility of 28Mn5Ni alloy are nearly similar to that of 28Mn0Ni alloy. However, there are marked differences in hardness and tensile behaviour of 28Mn0Ni and 15Mn5Ni alloys, with the latter being stronger but less ductile. The differences in hardness and yield strength values between these two alloys are about HV120 and 450 MPa, respectively. The 15Mn5Ni alloy displayed just 20% total elongation to failure against 55–68% by 28Mn(0,5)Ni alloys (Table 3). The microstructural evolution and response to deformation in terms of strength, strain hardening and ductility are discussed separately in the following sub-sections.

4.2.1. Microstructural evolution in LDS alloys during tensile deformation

The results show that the evolution of microstructure in LDS alloys with coarse B2 (15Mn5Ni) and B2-free (28Mn0Ni) alloys are markedly different with the tensile strains. Based on the observations, a general schematic for the microstructural evolution is illustrated in Fig. 11. In the B2-free alloy (28Mn0Ni), the initial microstructure after hot rolling consists of austenite grains with a few annealing twins and fine κ-carbides (Fig. 11a). Fig. 11e represents the initial microstructure, showing κ-carbides within γ-grains, B2 stringers connected across several γ-grains and sparse fine B2 along grain boundaries of B2 forming alloy (15Mn5Ni) in the hot-rolled condition. When subjected to tensile deformation, the B2-free alloy exhibits planar dislocation glide and pinning (Fig. 11b). At similar strains, the alloy with B2 shows substantial dislocation activity within γ-grains containing carbides and B2 grains. However, the grains in a stringer with a favourable orientation relationship with neighbouring austenite (marked ‘f’) undergo slip transfer across the γ/B2 interfaces, while non-favourable/hard B2 grains (marked as ‘nf’) show dislocation pile-up at the γ/B2 interfaces (Fig. 11f).

With progressive deformation, the B2-free alloy shows slip band initiation within γ-grains along with the slip-twin interaction across annealing twin boundaries (Fig. 11c). The γ-grains in B2-containing alloy exhibit slip band formation, deformation twinning along with dislocation interaction with coarse near cuboidal-shaped κ-carbides by particle shearing and dislocations pile-up near fine B2 particles (Fig. 11g). The grains that exhibited deformation twinning did not contain any slip bands. The B2 stringers in this alloy showed increasing slip transfer

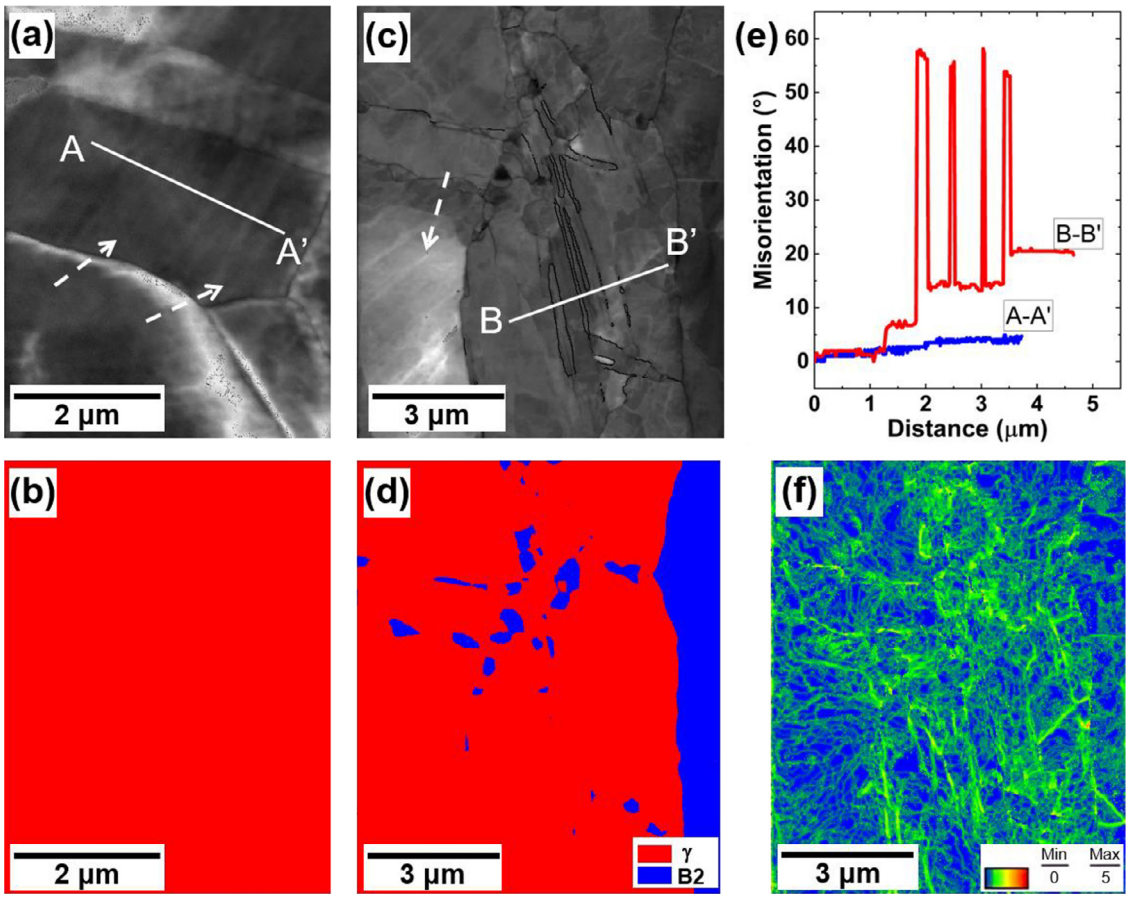


Fig. 9.. ACOM virtual bright-field micrographs and phase maps of (a,b) 28Mn0Ni and (c,d) 15Mn5Ni alloys deformed up to $\varepsilon \sim 0.15$, respectively. (e) Misorientation profile across AA' line shown in Fig. 9a and BB' line in Fig. 9c. (f) KAM map showing strain distribution inside the grains and interface surrounded by B2 in the 15Mn5Ni alloy.

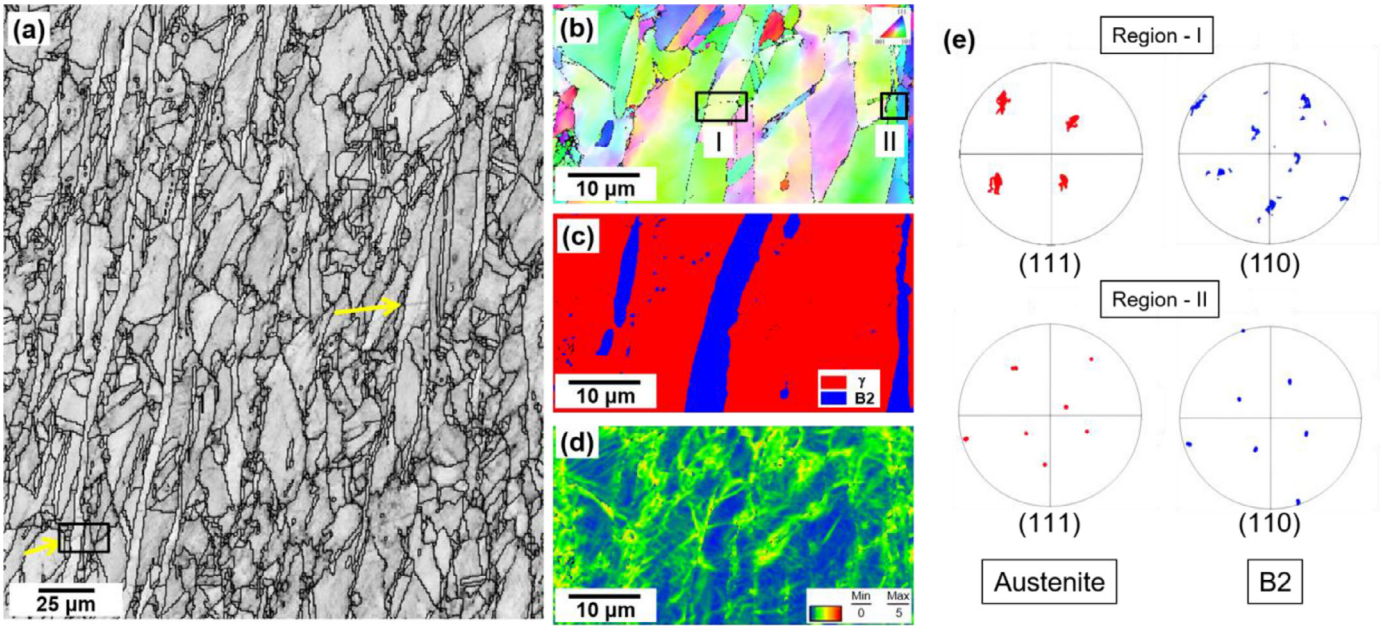


Fig. 10.. (a) EBSD-IQ map of cross section microstructure of tensile fractured 15Mn5Ni alloy, (b) magnified OIM map showing cracked B2, (c) phase map, (d) KAM map showing slip transfer and strain distribution in γ and B2. (e) Pole figures showing OR between γ and B2 in region I and II of Fig. 10b having cracked B2 and non-cracked B2, respectively.

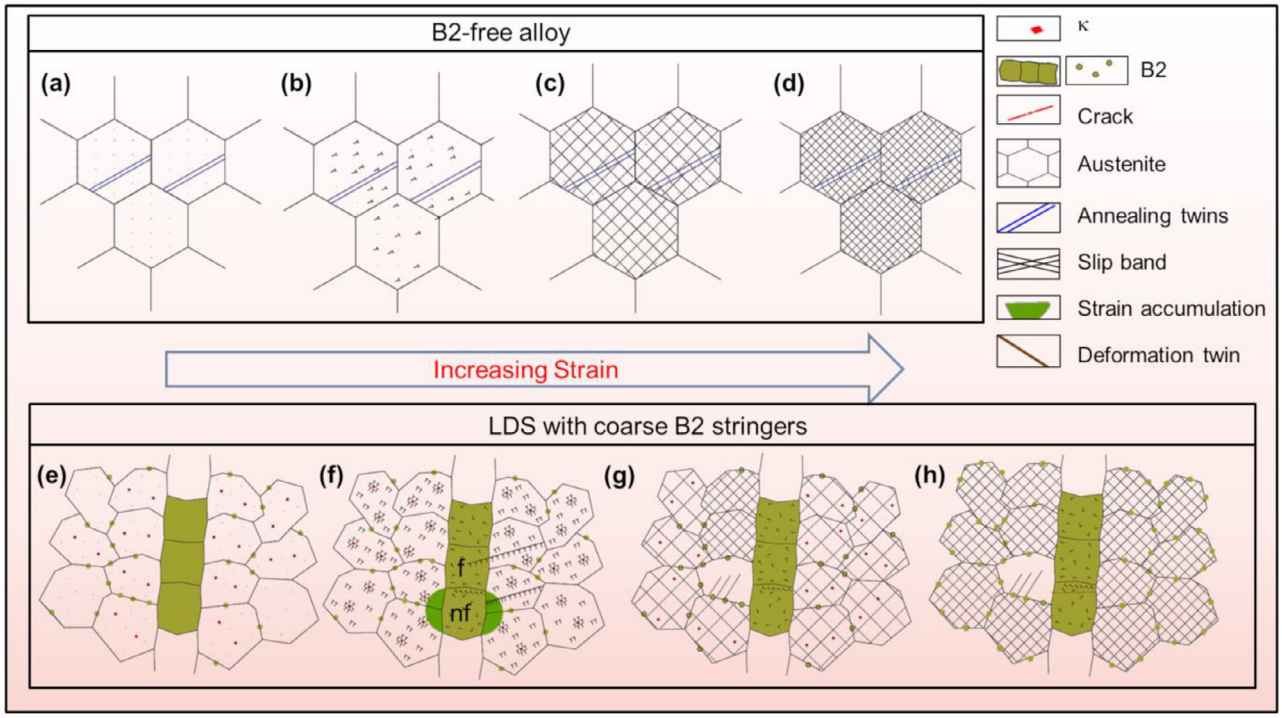


Fig. 11.. Schematic showing deformation mechanisms in (a-d) 28Mn0Ni and (e-h) 15Mn5Ni alloys with progress of deformation. (a) and (e) are initial microstructures of 28Mn0Ni and 15Mn5Ni alloys, respectively. (b, f), (c, g) and (d, h) are at low strain near yield, intermediate strain and fracture, respectively. Here, 'f' and 'nf' represent favourable and non-favourable OR of B2 with γ, respectively.

across favourable interfaces and pile-up around non-favourable ones. The B2-free alloy exhibited slip band refinement and saturation of mean slip band spacing with subsequent deformation (Fig. 11d), while the B2 forming alloy showed inhomogeneous slip band refinement and cracking of B2 at higher strains (Fig. 11h).

4.2.2. Strength

The major factors which contribute to the overall strength of LDS alloys are (i) strength of both the recrystallised and the deformed austenite, (ii) strengthening due to κ-carbides, (iii) strengthening due to B2 phase (and DO_3 precipitates if any), and (iv) strain partitioning between the austenite and B2 stringers/particles. The yield strength of the recrystallised austenite can be calculated using the following expression [28,33]:

$$\sigma_{y,Rx-aust} = \sigma_0 + k_{HP} d^{-1/2}$$

Where σ_0 is lattice frictional stress, d is the average recrystallised grain size, and k_{HP} is the Hall-Petch coefficient. The values of σ_0 and K_{HP} were determined from the experimental results reported by Etienne et al. [33] for Fe-28.8Mn-9.4Al-0.92C and are 288 MPa and 461 MPa $\mu m^{1/2}$, respectively. The estimated yield strength of the recrystallised austenite having a grain size in the range of ~5–7 μm, as in present LDS alloys, is about 460–500 MPa. This value is about 170–200 MPa less than that of the experimentally observed yield strength of the 28Mn0Ni alloy. Additionally, the yield strength of 28Mn0Ni alloy is higher than those reported for alloys of similar composition under cold rolled plus annealed conditions [11,12,33]. This difference can be attributed to the volume fraction of κ-carbides.

Nanoindentation study reveals that both B2 and γ-matrix exhibit similar hardness values in the range of 6.7–6.8 GPa (Table 3). Rahnama et al. [25] investigated the nanomechanical response of the individual or the combined precipitates depending on their size along with the γ-matrix in Fe-15Mn-10Al-0.8C-(0,5)Ni LDS alloys. They reported the hardness values for $\gamma+\kappa$, γ+B2, $\gamma+\kappa$+B2, and B2 stringer as ~7 GPa, ~8 GPa, ~9 GPa and ~9–11 GPa, respectively. The average hardness of $\gamma+\kappa$ in the present LDS alloy is close to those reported by Rahnama et al. [25]. It is important to note that the B2 phase can exhibit a wide range of hardness due to the differences in nature of bonds such as Fe-Al and Ni-Al in Ni-free and Ni containing LDS alloys, respectively, and variation in the annealing/rolling temperatures. The similar hardness values of the γ-matrix and B2 in the present study implies that both γ and B2 phases can deform simultaneously. However, the load-depth curves of B2 stringer displayed a characteristic 'pop-in' event in the loading segment (inset in Fig. 4b). A pop-in event involves a sudden burst in displacement during the load-controlled experiment. These pop-in events were notably absent during the indentation of γ-grains. Pop-in events are reported to be linked to incipient plasticity, where an elastic-plastic transition occurs beneath the indenter [34]. The sharp bursts of activity result from thermally activated deformation processes such as dislocation nucleation or release of pinned dislocations. A recent study on the nanoindentation of dual-phase steel under different heat treatments highlighted the effect of carbon diffusion-induced dislocation pinning, leading to similar pop-in behaviour as observed in the current study [35]. The possibility of pop-ins resulting from the release of pinned dislocations thus presents an interesting opportunity for study as the compositional differences between B2 and austenite were quantified without carbon. Also, presence of DO_3 phase was also observed in the B2 stringers (Fig. 3d). This difference in carbon content or the DO_3 phase in B2 could contribute to the initial dislocation pinning. Li et al. [36] investigated the pop-in nature in B2 (NiAl) single crystals and reported strong crystallographic dependence of the first pop-in loads with lowest for ⟨111⟩ and highest for ⟨001⟩ orientations. Hence, the range of first pop-in loads (~50–200 μN) noticed in B2 grains (Fig. 4b) could be a resultant of the orientation dependence. Currently, this is out of the scope of this work and it requires further experiments for validation.

The evidence of slip transfer occurring across favourable orientations (Fig. 7c) in 15Mn5Ni confirms both the γ-matrix and B2 would deform together as composite yielding occurs. The higher yield strength of 15Mn5Ni alloy as compared to 28Mn0Ni (and 28Mn5Ni) can be at-

tributed to (i) the presence of a large fraction of relatively coarser κ-carbides within the γ-matrix (Fig. 3g), (ii) a bamboo-like grain morphology of B2 stringers containing fine DO_3 ordered precipitates and (iii) the restricted slip activity across non-favourable γ/B2 interfaces. The increase in κ particles size and fraction changes the deformation mechanism from weakly coupled particle shearing to strongly coupled one and eventually to Orowan looping, as suggested by Yao et al. [31]. This transition in mechanisms can be expected in an alloy when the size of κ-carbides is ~20 nm [31]. With a wide size distribution of κ particles in the range of 5–40 nm in the present LDS alloys, either particle pining of dislocations (Fig. 7a), shearing (Fig. 8c) or looping (Fig. 7b) can be expected (Fig. 7b). Non-shearable B2 particles can also increase the strength of the alloy further due to dislocation pile-up around them (Figs. 7h, 8e & 9f) [10,28].

4.2.3. Strain hardening and ductility

The role of B2 phase on the strain hardening behaviour of LDS alloys has become a debatable topic in the recent past, and as highlighted in introduction section there exists two opposing view-points [21][22]. The mechanism proposed by Yang et al. [21] was based on the strain and stress partitioning between γ and B2 phases. They reported that deformation of B2 phase starts at a later stage of deformation ($\varepsilon \sim 0.08$) but strain hardens faster than γ phase. The transient plateau in the flow curve was thus attributed to this composite deformation. Subsequent back stress generation and forest hardening in the γ-matrix in the presence of B2 led to higher hardening rate observed in the B2 containing alloy. On the contrary, Kim [22] emphasised that the dislocation substructure evolution in austenite is governed by alloy composition, which plays a primary role, and B2 has a secondary role in strain hardening. Since there was no plateau in the stress-strain curve, Kim [22] also proposed that the transient portion of the flow curve does not originate due to the stress/strain partitioning; rather, it may be an indication of Lüder band formation. The absence of any noticeable plateaus in the stress-strain curves of all the three alloys (Fig. 5a) supports the argument given by Kim [22].

The slip activity in LDS alloy depends on the nature of γ/B2 interface. Favourably orientated soft B2 grains (with γ-grains) present in stringers can deform simultaneously through slip transfer near yield, whereas, slip transfer restriction across non-favourable γ/B2 interface increases the yield strength and affects subsequent strain hardening behaviour. It was observed that the strain hardening of both the alloys 28Mn0Ni and 15Mn5Ni mostly proceeds by planar glide and associated slip band refinement (Fig. 8). Similar dynamic slip band refinement has been reported in recent studies on B2-free alloys [18,20,31]. Planar glide of dislocation in B2-free alloys can be attributed to the glide plane softening caused by either shearing of κ-carbides or interaction between dislocations along with short-range order (SRO) [18,20,37]. The slip band spacing measured in the present study at a particular strain in 28Mn0Ni alloy is comparable to that reported by Welsch et al. [20]. Although the deformation substructure evolution and associated strain hardening have been reported for B2-free alloy, the same in the γ-matrix in presence of B2 has not been studied in-detail. The slip band spacing measured at a given strain ($\varepsilon \sim 0.15$) in 28Mn0Ni alloy is ~100–200 nm whereas in 15Mn5Ni alloy it is ~400–600 nm. Moreover, the slip bands are homogeneously distributed in the 28Mn0Ni alloy as compared to in 15Mn5Ni alloy.

Slip band nucleation occurs when the critical resolved shear stress exceeds the overall stress required to generate a Frank-Read source for a given loop length and the friction stress. Once a slip band nucleates, there will be a significant increase in dislocation density due to restricted cross-slip activity. With further deformation, more dislocations can pile-up within bands, exhaust the active source due to the force equilibrium reached between the applied stress and the mutual repulsive stress between dislocations. A new source must be initiated to accommodate further strain, thereby continuing the deformation process. This process leads to dynamic slip band refinement. The slip band spacing also depends on the long-range stress field between bands. The nature of these stress fields is governed by the number of dislocations piled-up within a band [38]. The presence of coherent precipitates in the matrix can also affect the stress fields and subsequently influence the microstructural evolution according to their size and distribution. Kuhlmann-Wilsdorf proposed that the dislocations are anchored in the presence of precipitates, resulting in a decrease in dislocation loop length which reduces the number of active sources at given stress and thereby affecting the slip band spacing [39]. Hence, it is expected that the slip bands can be refined at a slower rate in the presence of non-shearable particles, whereas shearable particles can enhance the slip band refinement rate. The current study observed κ-carbides in 28Mn0Ni alloy to be smaller and lesser in volume fraction and are sheared early during deformation. This resulted in the observed finer slip band spacing as compared to the 15Mn5Ni alloy. It can be pointed out that the κ-carbides in 15Mn5Ni alloy are not fully sheared across the specimen at low strain due to their inhomogeneous distribution which result in non-uniform slip band spacing. Apart from the κ-carbides, non-shearable B2 particles in the 15Mn5Ni alloy can also affect the slip band refinement. In the presence of B2, the active slip length reduced further, thereby leading to high yield stress and contributing to a slower rate of slip band refinement. Thus, the overall slip band spacing in 15Mn5Ni alloy decreases at a much slower rate, leading to a lower strain hardening rate than in 28Mn0Ni alloy. Apart from dynamic slip band refinement, annealing twin-slip interaction is also prominent in 28Mn0Ni alloy. Thus, the differences in strain-hardening characteristics of 28Mn0Ni and 15Mn5Ni alloy can be attributed to various initial microstructural features such as κ-carbides (its size and volume fraction), B2 (type, morphology and orientation relationship with the matrix) and subsequent dislocation interactions in the presence of these phases during deformation.

It is well-understood that the strain hardening exponent (n) is a measure of uniform strain that a polycrystalline material can accommodate. As noted in Fig. 5b, 15Mn5Ni alloy displayed a lower 'n' value than 28Mn(0,5)Ni alloys, resulting in different ductility levels. The fracture surface of the failed tensile specimen of 15Mn5Ni alloy shows cleavage facets (in some B2 phase) connected by dimple morphology (Fig. 6c) and is consistent with that reported by Park et al. [23]. The microstructure of the failed specimen also showed cracks in B2 stringers (Fig. 10a,b), implying that the slip transfer at the γ/B2 interface plays a vital role in the composite deformation behaviour of 15Mn5Ni alloy. The deformation behaviour is also influenced by grain-to-grain slip transfer within the polycrystalline B2 stringer, as illustrated schematically in Fig. 11e-h. There will be substantial strain accumulation around hard orientated B2 while an easy slip transfer occurs in the case of soft orientation. Furthermore, the soft orientated B2 deforms in conjunction with the adjacent matrix and its texture evolves to maintain the orientation relationship. The strain accumulation observed around the neighbouring hard-orientated B2 results in deformation heterogeneity (Fig. 7h and Fig. 9f) and subsequent crack nucleation in B2 (Fig. 10). On further straining, the inhibition of slip transfer near γ/B2 interfaces leads to higher local misorientation within a γ grain. These re-orientated regions can also facilitate deformation twinning (Fig. 8f) due to the medium SFE of the alloy as reported by Abedi et al. [40]. In summary, the nature of γ/B2 interfaces (non-favourable OR) and dislocation accommodation within γ and B2 influence the strain hardening and the overall ductility of LDS alloys.

5. Conclusions

Three different LDS grades viz. Fe-28Mn-9Al-0.9C, Fe-28Mn-9Al-5Ni-0.9C and Fe-15Mn-9Al-5Ni-0.9C, were produced by vacuum induction melting and subsequently hot-rolled. Microstructures of all alloys consisted of a γ-matrix containing κ-carbides. An additional B2 phase seen to form in Ni containing LDS alloys. B2 was present in the form of long polycrystalline stringers and fine polygonal particles. The present study aimed to understand the role of B2 on the tensile response and mi-

crostructural evolution of these alloys. The key findings from the present study are as follows:

- Rich B2 phase containing alloy exhibited the highest strength with reduced ductility compared to the other alloys. The γ/B2 interface played a critical role in controlling the strength and ductility of B2 containing alloys. A favourable orientation relationship (OR) between the γ-matrix and B2 facilitated slip transfer while the non-favourable OR controlled the ductility by strain accumulation and B2-cracking.
- Localised nanoindentation of γ-matrix and grains in B2 stringer showed similar indentation response with B2 having a characteristic 'pop-in'. This implies that during yield, co-deformation of γ and B2 occurs along with slip transfer across favourable γ/B2 interfaces as confirmed by transmission electron microscopy.
- The presence of both κ-carbides and B2 modified the dislocation substructure evolution in austenite during deformation. Dislocations were accumulated around κ channels during yielding, which evolved further as slip bands at higher strains followed by dynamic slip band refinement. Slip band refinement in B2-free alloy was much faster with uniform slip band spacing of 100–200 nm, whereas the presence of B2 limited the slip band refinement. This led to coarse and inhomogeneous slip bands with spacing of 400–600 nm in 15Mn5Ni alloy.
- At later stages of deformation, slip restriction at γ/B2 interface resulted in high local re-orientation within a γ-grain surrounded by B2 particles/stringers in 15Mn5Ni alloy as compared to the B2-free 28Mn0Ni alloy. Deformation twinning was also noticed in such re-orientated grains which was absent in the B2-free 28Mn0Ni alloy.

The present study's findings provide a critical understanding to evaluate microstructural modifications to enhance the mechanical properties of B2 containing low density steels.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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